

Figure 1(a): Authentic Dogs vs. Cats Photographic Dataset Samples (Kaggle / Udacity Challenge)

Real photographic samples showing high natural variance (12,500 cats, 12,500 dogs, 25,000 total images - balanced 1:1 ratio)

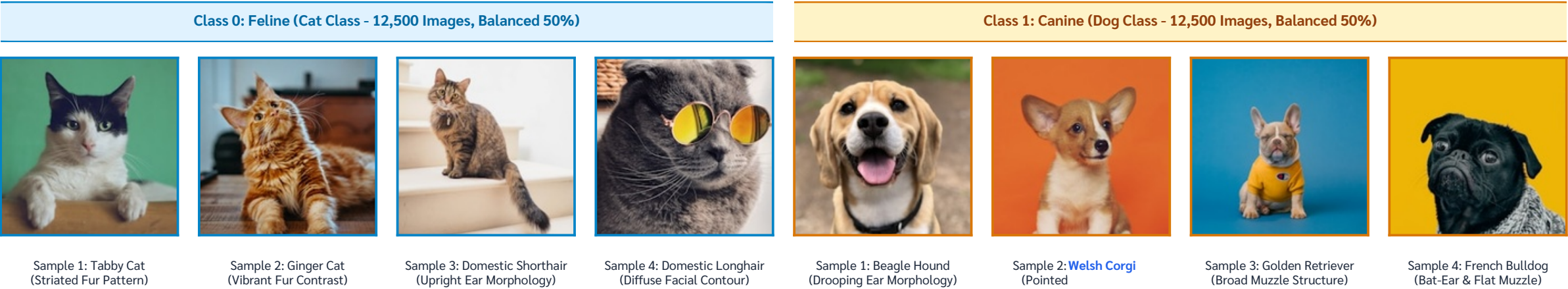


Figure 1(b): Dynamic Data Augmentation Pipeline on Real Photographs (Stochastic Transformation Flow)

Mini-batch stochastic transformations enforce affine and photometric invariance, regularizing against overfitting.

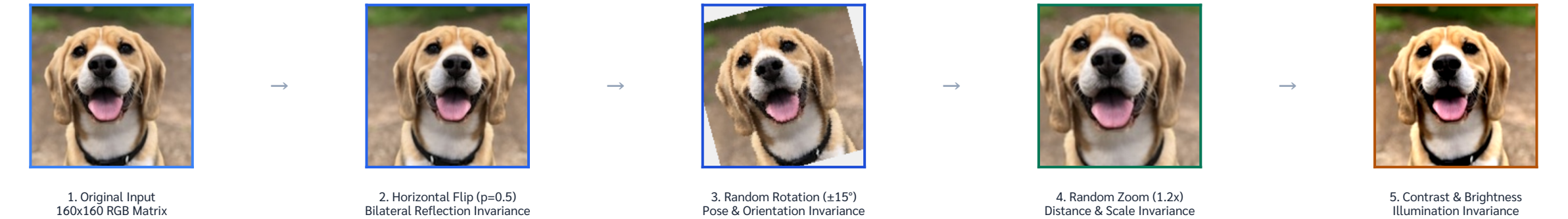


Figure 1(c): Pixel Intensity & RGB Color Channel Distribution Analysis (Class 0: Cat vs. Class 1: Dog)

Nearly identical pixel density curves confirm that raw intensities are non-separable, necessitating deep spatial convolution.

